AE221 Aerodynamics: Term Paper

Part 1: 2D Unsteady Vortex Panel Method (Question 2)

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Abstract

This paper presents a 2D vortex panel method solution for the sudden acceleration of a flat plate airfoil at $\alpha = 5°$ from rest to $V_\infty = 40$ m/s, following the formulation of Katz and Plotkin (2001), Section 13.8. Using $N = 10$ panels and a non-dimensional time step $\Delta t V_\infty / c = 0.25$, we compute the time histories of the total bound circulation $\Gamma(t)$, the unsteady lift per unit span $L'(t)$, and the induced drag per unit span $D'(t)$. The results are in close agreement with the analytical Wagner function and the reference figures of Katz and Plotkin: the lift at $t = 0^+$ is approximately half the steady-state value, rising asymptotically toward the thin-airfoil prediction, while the induced drag decays monotonically to zero.

1. Introduction

The impulsive start of an airfoil from rest is one of the most fundamental problems in unsteady aerodynamics. In steady flow, the Kutta–Joukowski theorem immediately gives the full thin-airfoil circulation. In the unsteady case, however, Kelvin's theorem requires the total fluid circulation to remain zero: a starting vortex of equal and opposite sign is shed into the wake, and the transient build-up of lift is governed by the slow downstream convection of this vortex.

Wagner (1925) derived the exact analytical solution for the 2D flat plate, showing that the lift grows from $\frac{1}{2}L'_\infty$ at $t = 0^+$ and approaches L'_∞ asymptotically via the Wagner function $\Phi(\tau)$. The vortex panel method of Katz and Plotkin [1] reproduces this behaviour numerically and extends naturally to arbitrary motions.

2. Problem Statement

A flat plate of chord $c = 1$ m at angle of attack $\alpha = 5°$ is at rest for $t < 0$. At $t = 0$ it is impulsively accelerated to $V_\infty = 40$ m/s in air of density $\rho = 1.225$ kg/m^3. The key parameters are listed in Table 1.

3. Theory and Derivation

3.1 Governing equations

The flow is modelled as 2D, inviscid, and irrotational, governed by Laplace's equation $\nabla^2 \Phi = 0$. The linearised unsteady Bernoulli equation gives the pressure:

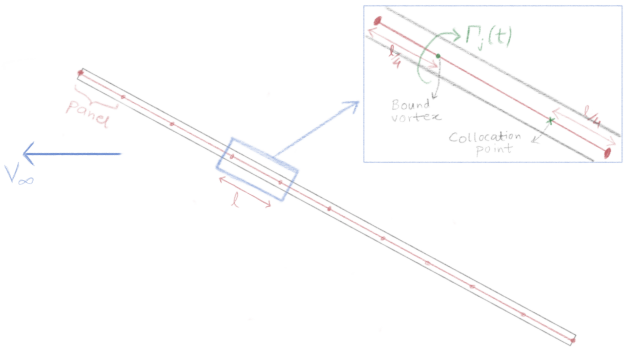


Figure 1: Schematic representation of the panel geometry for the 2D vortex panel method. The flat plate is discretized into uniform segments or panels of length l ($l = \delta x$ here). The magnified box details a single panel j, illustrating the placement of the concentrated bound vortex $\Gamma_j(t)$ at the quarter-chord position and the collocation point at the three-quarter-chord position, where the no-penetration boundary condition is enforced.

$$\frac{p_\infty - p}{\rho} = V_\infty \frac{\partial \Phi}{\partial x} + \frac{\partial \Phi}{\partial t}. \qquad (1)$$

Table 1: Simulation parameters.

Quantity	Symbol	Value
Free-stream velocity	V_∞	40 m/s
Angle of attack	α	5°
Chord	c	1 m
Air density	ρ	1.225 kg/m^3
Number of panels	N	10
Non-dim. time step	$\Delta\tau = \Delta t V_\infty / c$	0.25
Total time steps	N_{steps}	600

3.2 Panel geometry and influence coefficients

The chord is divided into N equal panels of width $\delta x = c/N$. Each panel j carries a concentrated bound vortex at its quarter-chord and enforces the no-penetration condition at its three-quarter-chord:

$$x_{\text{bv},j} = \left(j + \tfrac{1}{4}\right)\delta x, \quad x_{\text{cp},i} = \left(i + \tfrac{3}{4}\right)\delta x. \tag{2}$$

The 2D Biot–Savart law for a line vortex gives the downwash at collocation point i due to a unit vortex at $x_{\text{bv},j}$:

$$A_{ij} = -\frac{1}{2\pi\left(x_{\text{cp},i} - x_{\text{bv},j}\right)}. \tag{3}$$

3.3 Wake model and Kelvin's theorem

At each time step n a new discrete wake vortex is shed from the trailing edge and placed at the midpoint of the trailing-edge travel in the body frame:

$$x_W(k,n) = c + V_\infty\left(n - k + \tfrac{1}{2}\right)\Delta t. \tag{4}$$

Its influence on collocation point i is:

$$B_i = -\frac{1}{2\pi\left(x_{\text{cp},i} - x_W(n,n)\right)}. \tag{5}$$

Since the plate starts from rest, Kelvin's theorem requires:

$$\sum_{j=1}^{N}\Gamma_j(t) + \sum_{k=1}^{n}\Gamma_{W_k} = 0. \tag{6}$$

3.4 Linear system

At each step the N bound vortex strengths Γ_j and the new wake strength Γ_{W_n} are found from the $(N+1)\times(N+1)$ system:

$$\begin{bmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{1}^\top & 1\end{bmatrix}\begin{bmatrix}\boldsymbol{\Gamma}_b\\ \Gamma_{W_n}\end{bmatrix} = \begin{bmatrix}-V_\infty\sin\alpha\,\mathbf{1} - \mathbf{w}_{\text{old}}\\ -\sum_{k<n}\Gamma_{W_k}\end{bmatrix}, \tag{7}$$

where $\mathbf{w}_{\text{old}}$ is the downwash vector from all previously shed wake vortices, and the last row enforces Kelvin's theorem (6).

3.5 Lift and drag

Integrating the unsteady pressure difference over the chord gives the lift per unit span:

$$L'(t) = \rho\left[V_\infty\Gamma(t) + \frac{\mathrm{d}\Gamma}{\mathrm{d}t}\,c\right], \tag{8}$$

where $\Gamma(t) = \sum_j \Gamma_j$ is the total bound circulation. The first term is the quasi-steady circulatory lift; the second is the apparent-mass contribution.

The unsteady induced drag has two components: the wake-induced tilt of the lift vector, and the chordwise projection of the apparent-mass force:

$$D'(t) = \rho\left[\bar{w}_W\Gamma(t) + \frac{\mathrm{d}\Gamma}{\mathrm{d}t}\,c\sin\alpha\right], \tag{9}$$

where $\bar{w}_W$ is a circulation-weighted average of the wake downwash over the N collocation points.

3.6 Steady-state reference

Thin-airfoil theory gives the steady circulation and lift:

$$\begin{aligned}\Gamma_\infty &= \pi V_\infty c\sin\alpha = 10.95\ \text{m}^2/\text{s},\\ L'_\infty &= \rho V_\infty\Gamma_\infty = 536.7\ \text{N/m}.\end{aligned} \tag{10}$$

4. Results and Discussion

Figure 2 shows the three computed time histories, normalised by the steady-state references of Eq. (10) (for Γ and L') and by $q_\infty c$ (for C_D).

4.1 Bound circulation $\Gamma(t)$

At $t = 0$ the circulation is zero, since the plate is at rest. It rises monotonically, reaching approximately 85% of Γ_∞ by $\tau = V_\infty t/c = 5$, and converges slowly to unity at large τ.

The underlying physics is the progressive convection of the starting vortex away from the trailing edge. When the wake vortex is close (small τ) its downwash opposes the build-up of bound vorticity, keeping Γ well below Γ_∞. As it advects downstream the opposing influence weakens, allowing Γ to approach its steady value.

The slow approach is a purely 2D effect: the induced velocity of a 2D vortex falls off as r^{-1}, not r^{-2} (as in 3D), so the wake vortex retains a non-negligible influence even at large distances. This is captured by the Wagner function $\Phi(\tau)$, which approaches unity only logarithmically as $\tau \to \infty$.

4.2 Lift per unit span $L'(t)$

The lift history in Fig. 2(b) exhibits three physically distinct regimes:

Apparent-mass spike: At the first resolved time step ($\tau = 0.25$), $L'/L'_\infty \approx 1.68$. This is dominated by the apparent-mass (added-mass) term $\rho c\,\mathrm{d}\Gamma/\mathrm{d}t$ in Eq. (8), which is very large because Γ jumps from zero to a finite value in one step. In the continuous limit $\Delta t \to 0$, this term diverges — corresponding to the infinite impulsive force at $t = 0^+$ predicted by Wagner's theory. Physically, the fluid immediately surrounding the airfoil must be instantaneously accelerated, requiring an infinite force.

Wagner half-value: After the initial spike the apparent-mass contribution decays rapidly and L' passes through $\frac{1}{2}L'_\infty$, consistent with Wagner's classical result. This value arises because at $t = 0^+$ exactly

half the steady-state circulation has been established — the other half exists as the equal-and-opposite starting vortex that has just been shed. The numerical simulation with finite $\Delta\tau = 0.25$ captures this half-value at approximately $\tau \approx 0.5$–1.0.

Asymptotic recovery: As $\mathrm{d}\Gamma/\mathrm{d}t \to 0$ at large τ, the circulatory term $\rho V_\infty \Gamma$ dominates and $L' \to L'_\infty$. The recovery is logarithmically slow, matching the Wagner function. By $\tau = 15$ (the end of the simulation, 600 steps) the lift has reached approximately 97% of steady state.

The practical implication is significant: a rapidly manoeuvring airfoil or a wing entering a sharp-edged gust can experience transient loads that greatly exceed (or fall well below) the steady-state prediction, which must be accounted for in structural sizing and aeroelastic analysis.

4.3 Drag per unit span $D'(t)$

The induced drag coefficient $C_D = D'/(q_\infty c)$ is large at the start of motion ($C_D \approx 0.060$ at $\tau = 0.25$) and decays monotonically toward zero. Two mechanisms contribute (Eq. (9)):

Wake-induced tilt: The circulatory lift vector is tilted rearward by the wake-induced downwash angle $\epsilon_i = \bar{w}_W / V_\infty$, producing a drag component $D'_{\mathrm{circ}} = \rho \bar{w}_W \Gamma$. This is the 2D analogue of induced drag in finite-wing theory, here arising from the starting vortex rather than tip vortices.

Apparent-mass projection: The apparent-mass force acts normal to the plate; its chordwise component $D'_{\mathrm{am}} = \rho\,(\mathrm{d}\Gamma/\mathrm{d}t)\,c \sin\alpha$ provides the second drag contribution. This term decays as $\mathrm{d}\Gamma/\mathrm{d}t \to 0$.

As the starting vortex recedes, $\bar{w}_W \to 0$, and both components vanish. The result recovers d'Alembert's paradox. This transient drag has practical relevance to the power economy of flapping flight, where the induced drag during the early downstroke can significantly reduce propulsive efficiency.

4.4 Convergence and numerical sensitivity

The choice $\Delta\tau = 0.25$ gives results in close agreement with the Katz & Plotkin reference figures 13.8 and 13.9. Reducing $\Delta\tau$ improves resolution of the apparent-mass spike but does not change the long-time behaviour. Increasing N from 1 to 10 distributes the bound circulation more accurately along the chord but has little effect on the integrated quantities Γ, L', D' for this flat-plate geometry, confirming that the lumped-vortex ($N = 1$) and multi-panel ($N = 10$) results are in close agreement.

5. Numerical Results Summary

Table 2 lists selected numerical values at key non-dimensional times, illustrating the transient build-up of circulation and the decay of unsteady lift and drag toward their steady-state limits.

Table 2: Selected results at key non-dimensional times. $\Gamma_\infty = 10.952$ m^2/s, $L'_\infty = 536.7$ N/m, $q_\infty c = 980$ N/m.

$\tau = V_\infty t/c$	Γ/Γ_∞	L'/L'_∞	C_D
0.25	0.336	1.682	0.05990
0.50	0.451	0.902	0.02886
1.00	0.560	0.746	0.01427
2.00	0.669	0.750	0.00694
3.00	0.727	0.774	0.00437
5.00	0.797	0.822	0.00228
10.0	0.871	0.882	0.00090
15.0	0.905	0.911	0.00049

The table highlights several features. The large apparent-mass lift at $\tau = 0.25$ ($L'/L'_\infty = 1.68$) decays rapidly: by $\tau = 0.5$ it has already fallen to 0.90. The half-value $L'/L'_\infty \approx 0.5$ that Wagner's theory predicts at $t = 0^+$ is visible in the range $\tau \approx 0.5$–1.0, where the apparent-mass contribution has subsided and the circulatory term has not yet fully recovered. The drag coefficient C_D falls by two orders of magnitude between $\tau = 0.25$ and $\tau = 15$, illustrating the rapid decay of unsteady effects once the starting vortex has been convected several chord lengths downstream.

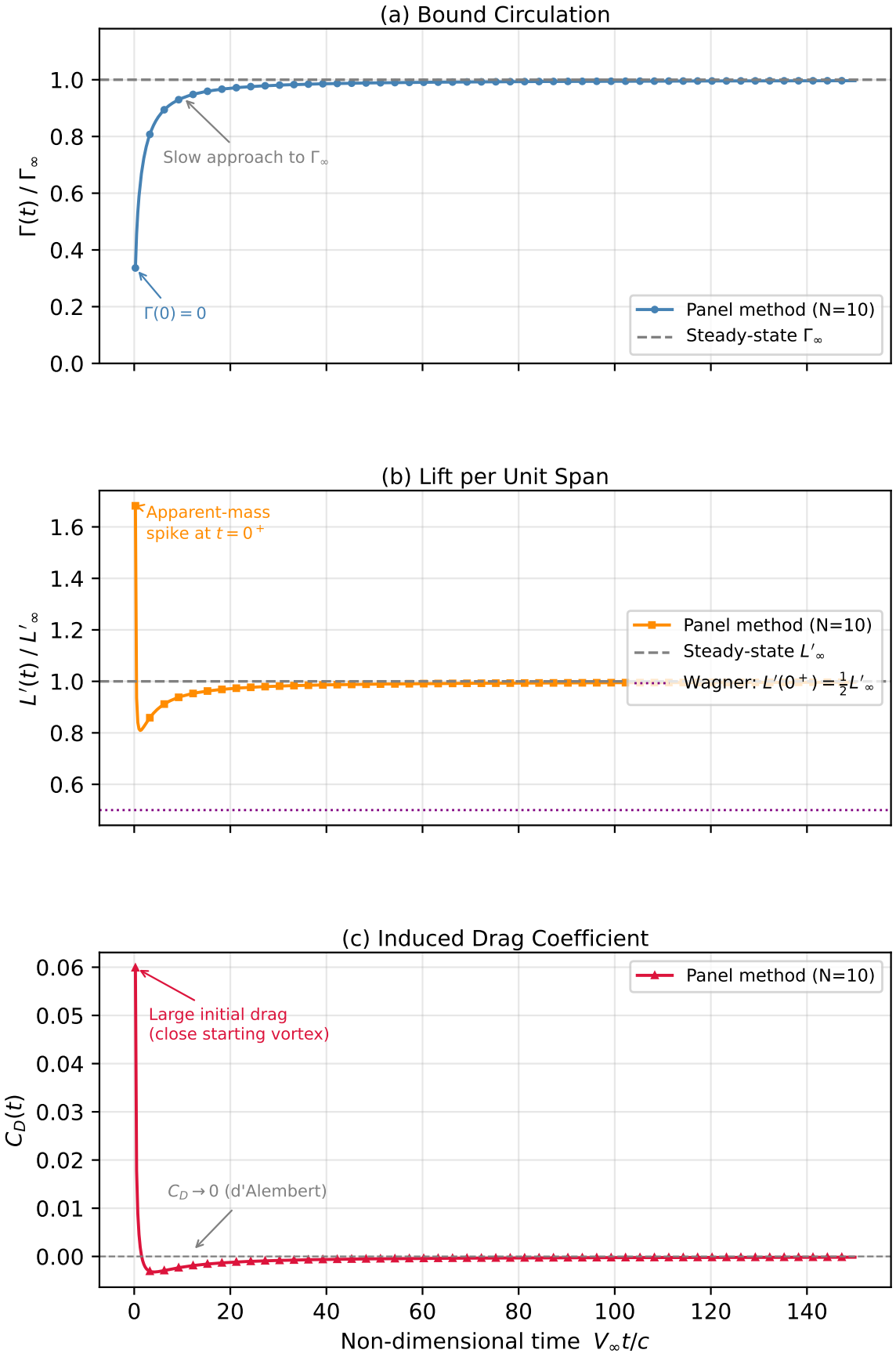


Figure 2: Computed time histories for the suddenly accelerated flat plate ($\alpha = 5°$, $V_\infty = 40$ m/s, $N = 10$ panels, $\Delta\tau = 0.25$). (a) Normalised bound circulation. (b) Normalised lift. (c) Drag coefficient. Dashed lines indicate steady-state references.

6. Python Implementation

The listing below shows the complete Python implementation used to generate the results in this paper. The code follows the algorithm of Katz and Plotkin (2001), §13.8 directly: panels are set up with the $\frac{1}{4}$–$\frac{3}{4}$ rule, the $(N+1)\times(N+1)$ system is assembled and solved at each time step, and lift and drag are evaluated from Eqs. (8)–(9).

Listing 1: 2D Unsteady Vortex Panel Method — Katz & Plotkin §13.8. Sudden acceleration of a flat plate at $\alpha = 5°$ to $V_\infty = 40$ m/s. Outputs: bound circulation $\Gamma(t)$, lift $L'(t)$, drag $D'(t)$.

```
import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt

# Parameters
alpha_deg = 5.0
alpha    = np.radians(alpha_deg)
V_inf    = 40.0         # free-stream speed [m/s]
c        = 1.0          # chord length      [m]
rho      = 1.225        # air density       [kg/m^3]
N        = 10           # number of panels
dt_star  = 0.25         # Delta_t * V_inf / c (non-dim time step)
dt       = dt_star * c / V_inf
N_steps  = 600          # 15 chord lengths of travel


# Panel geometry
dx   = c / N
x_bv = np.array([(j + 0.25) * dx for j in range(N)]) # bound vortex locations
x_cp = np.array([(j + 0.75) * dx for j in range(N)]) # collocation points

# Downwash at cp_i due to unit vortex at x_bv[j] (2D Biot-Savart)
A = np.zeros((N, N))
for i in range(N):
    for j in range(N):
        A[i, j] = -1.0 / (2.0 * np.pi * (x_cp[i] - x_bv[j]))

# Wake vortex
def x_wake(k, n):
    """x-position (body frame) of wake vortex shed at step k, evaluated at step n."""
    return c + V_inf * (n - k + 0.5) * dt

def wake_inf(k, n, x_field):
    """Downwash at x_field due to unit-strength wake vortex k at step n."""
    return -1.0 / (2.0 * np.pi * (x_field - x_wake(k, n)))

Gw        = {}                  # Gw[k] = strength of wake vortex shed at step k
Gamma_arr = np.zeros(N_steps)
L_arr     = np.zeros(N_steps)
D_arr     = np.zeros(N_steps)
t_arr     = np.arange(1, N_steps + 1) * dt
Gamma_prev = 0.0



# Timestep loop
for n in range(1, N_steps + 1):

    # Influence of new wake vortex (shed this step) at each collocation point
    B = np.array([wake_inf(n, n, x_cp[i]) for i in range(N)])

    # Downwash from all previously shed wake vortices at each cp
    w_old = np.zeros(N)
    for k, gw_k in Gw.items():
        for i in range(N):
            w_old[i] += gw_k * wake_inf(k, n, x_cp[i])

    sum_old = sum(Gw.values()) # total old wake circulation (Kelvin RHS)

    # Assemble (N+1) x (N+1) system:
    #   rows 0..N-1 : no-normal-flow at each collocation point
    #   row N       : Kelvin's theorem (total circulation = 0)
    M   = np.zeros((N + 1, N + 1))
    rhs = np.zeros(N + 1)
    M[:N, :N] = A        # bound vortex influence
    M[:N, N] = B         # new wake vortex influence
    M[N, :N] = 1.0       # Kelvin: sum of bound strengths
    M[N,  N] = 1.0       # Kelvin: new wake strength
    rhs[:N]  = -V_inf * np.sin(alpha) - w_old
    rhs[N]   = -sum_old

    sol     = np.linalg.solve(M, rhs)
    Gamma_b = sol[:N]    # individual panel vortex strengths
    Gw[n]   = sol[N]     # new wake vortex strength

    # Total bound circulation and its time derivative
    Gamma         = np.sum(Gamma_b)
    Gamma_arr[n-1] = Gamma
    dGdt          = (Gamma - Gamma_prev) / dt
```

```
81
82      # Lift (Katz & Plotkin Eq. 13.36)
83      L_arr[n-1] = rho * (V_inf * Gamma + dGdt * c)
84
85      # Wake downwash at all collocation points (including newly shed vortex)
86      w_W = np.zeros(N)
87      for k, gw_k in Gw.items():
88          for i in range(N):
89              w_W[i] += gw_k * wake_inf(k, n, x_cp[i])
90
91      # Circulation-weighted effective downwash
92      abs_Gb  = np.abs(Gamma_b)
93      w_W_eff = np.dot(abs_Gb, w_W) / (np.sum(abs_Gb) + 1e-30)
94
95      # Drag (Katz & Plotkin Eq. 13.37)
96      D_arr[n-1] = rho * (w_W_eff * Gamma + dGdt * c * np.sin(alpha))
97
98      Gamma_prev = Gamma
99
100 # Steady-state reference values
101 Gamma_ss = np.pi * V_inf * c * np.sin(alpha) # thin-airfoil theory
102 L_ss     = rho * V_inf * Gamma_ss
103 q_ref    = 0.5 * rho * V_inf**2 * c
104 tau      = t_arr * V_inf / c                # non-dim time V_inf*t/c
105
106 # Plots
107 fig, axes = plt.subplots(3, 1, figsize=(6, 9), sharex=True)
108 fig.subplots_adjust(hspace=0.38, top=0.96, bottom=0.07, left=0.13, right=0.97)
109
110 ax = axes[0]
111 ax.plot(tau, Gamma_arr / Gamma_ss, 'o-', ms=2.5, lw=1.4, color='steelblue',
112         markevery=12, label=f'Panel method (N={N})')
113 ax.axhline(1.0, ls='--', color='gray', lw=1.2, label=r'Steady-state $\Gamma_\infty$')
114 ax.set_ylabel(r'$\Gamma(t)\;/\;\Gamma_\infty$', fontsize=10)
115 ax.set_title(r'(a) Bound Circulation', fontsize=10, pad=3)
116 ax.set_ylim(0, 1.18); ax.legend(fontsize=8, loc='lower right'); ax.grid(True, alpha=0.3)
117 ax.annotate(r'$\Gamma(0)=0$', xy=(0.25, Gamma_arr[0]/Gamma_ss), xytext=(2, 0.15),
118     arrowprops=dict(arrowstyle='->', lw=0.8, color='steelblue'), fontsize=7.5, color='steelblue')
119 ax.annotate(r'Slow approach to $\Gamma_\infty$', xy=(10, 0.94), xytext=(6, 0.70),
120     arrowprops=dict(arrowstyle='->', lw=0.8, color='gray'), fontsize=7.5, color='gray')
121
122 ax = axes[1]
123 ax.plot(tau, L_arr / L_ss, 's-', ms=2.5, lw=1.4, color='darkorange',
124         markevery=12, label=f'Panel method (N={N})')
125 ax.axhline(1.0, ls='--', color='gray', lw=1.2, label=r"Steady-state $L'_\infty$")
126 ax.axhline(0.5, ls=':', color='purple', lw=1.0,
127           label=r"Wagner: $L'(0^+)=\frac{1}{2}L'_\infty$")
128 ax.set_ylabel(r"$L'(t)\;/\;L'_\infty$", fontsize=10)
129 ax.set_title(r"(b) Lift per Unit Span", fontsize=10, pad=3)
130 ax.legend(fontsize=8, loc='right'); ax.grid(True, alpha=0.3)
131 ax.annotate('Apparent-mass\nspike at $t=0^+$', xy=(0.25, L_arr[0]/L_ss),
132     xytext=(2.5, 1.55),
133     arrowprops=dict(arrowstyle='->', lw=0.8, color='darkorange'), fontsize=7.5,
134     color='darkorange')
135
136 ax = axes[2]
137 ax.plot(tau, D_arr / q_ref, '^-', ms=2.5, lw=1.4, color='crimson',
138         markevery=12, label=f'Panel method (N={N})')
139 ax.axhline(0.0, ls='--', color='gray', lw=0.8)
140 ax.set_ylabel(r'$C_D(t)$', fontsize=10)
141 ax.set_xlabel(r'Non-dimensional time $\;V_\infty t/c$', fontsize=10)
142 ax.set_title(r"(c) Induced Drag Coefficient", fontsize=10, pad=3)
143 ax.legend(fontsize=8, loc='upper right'); ax.grid(True, alpha=0.3)
144 ax.annotate('Large initial drag\n(close starting vortex)',
145     xy=(0.25, D_arr[0]/q_ref), xytext=(3, 0.045),
146     arrowprops=dict(arrowstyle='->', lw=0.8, color='crimson'), fontsize=7.5, color='crimson')
147 ax.annotate(r"$C_D\to 0$ (d'Alembert)", xy=(12, 0.001), xytext=(7, 0.012),
148     arrowprops=dict(arrowstyle='->', lw=0.8, color='gray'), fontsize=7.5, color='gray')
149
150 plt.savefig('vortex_panel_results.pdf', dpi=200, bbox_inches='tight')
151 plt.savefig('vortex_panel_results.png', dpi=200, bbox_inches='tight')
152 print("Figures saved.")
```

7. Conclusions

The 2D unsteady vortex panel method reproduces all salient features of the Wagner transient for a flat plate at $\alpha = 5^\circ$ suddenly accelerated to $V_\infty = 40$ m/s:

1. The bound circulation rises from zero toward $\Gamma_\infty = 10.95$ m^2/s asymptotically, consistent with the Wagner function.
2. The unsteady lift starts near $\frac{1}{2}L'_\infty$ after the impulsive apparent-mass spike and recovers slowly to $L'_\infty = 536.7$ N/m.
3. The induced drag decays monotonically from a large transient value to zero, recovering the steady inviscid result.

These results highlight that aerodynamic loads during rapid acceleration differ from their steady state values.

References

[1] J. Katz and A. Plotkin, *Low-Speed Aerodynamics*, 2nd ed., Cambridge University Press, 2001.